

Period 2

Ejercicio 1:

$$ds^2 - d\bar{s}_0^2 = \underbrace{(ds + d\bar{s}_0)}_{\approx 2 d\bar{s}_0} \cdot \underbrace{(ds - d\bar{s}_0)}_{d\Delta l} = 2 \underbrace{\epsilon_{ij}}_{\epsilon_{ij}} da_i da_j \Rightarrow$$

$$\Rightarrow d\Delta l = \epsilon_{ij} \frac{t_i}{d\bar{s}_0} \cdot \frac{t_j}{d\bar{s}_0} d\bar{s}_0$$

$$\int_{s_0} d\Delta l = \int_{s_0} \epsilon_{ij} t_i t_j d\bar{s}_0 \Rightarrow \Delta l = \int_{s_0} \epsilon_{ij} t_i t_j d\bar{s}_0$$

$$\Delta l_{\bar{a}\bar{b}} = \int_{\bar{a}\bar{b}} \epsilon_{ij} t_i t_j d\bar{s}_0 + \int_{\bar{b}\bar{c}} \epsilon_{ij} t_i t_j d\bar{s}_0 =$$

$$= \epsilon_{ij} t_i^{\bar{a}\bar{b}} t_j^{\bar{a}\bar{b}} \int_{\bar{a}\bar{b}} d\bar{s}_0 + \epsilon_{ij} t_i^{\bar{b}\bar{c}} t_j^{\bar{b}\bar{c}} \int_{\bar{b}\bar{c}} d\bar{s}_0 =$$

$$= \epsilon_{ij} t_i^{\bar{a}\bar{b}} t_j^{\bar{a}\bar{b}} l^{\bar{a}\bar{b}} + \epsilon_{ij} t_i^{\bar{b}\bar{c}} t_j^{\bar{b}\bar{c}} l^{\bar{b}\bar{c}} =$$

$$= \left(\frac{l^{\bar{a}\bar{b}} t_i^{\bar{a}\bar{b}}}{l^{\bar{a}\bar{b}}} \right) \epsilon_{ij} \left(\frac{l^{\bar{a}\bar{b}} t_j^{\bar{a}\bar{b}}}{l^{\bar{a}\bar{b}}} \right) + \left(\frac{l^{\bar{b}\bar{c}} t_i^{\bar{b}\bar{c}}}{l^{\bar{b}\bar{c}}} \right) \epsilon_{ij} \left(\frac{l^{\bar{b}\bar{c}} t_j^{\bar{b}\bar{c}}}{l^{\bar{b}\bar{c}}} \right)$$

$$l^{\bar{a}\bar{b}} \underline{t}^{\bar{a}\bar{b}} = \begin{bmatrix} \sqrt{3} \\ 1 \end{bmatrix} \Rightarrow l^{\bar{a}\bar{b}} = \sqrt{(\sqrt{3})^2 + 1^2} = 2$$

$$l^{\bar{b}\bar{c}} \underline{t}^{\bar{b}\bar{c}} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \Rightarrow l^{\bar{b}\bar{c}} = \sqrt{1^2 + 1^2} = \sqrt{2}$$

$$\underline{\epsilon} \cdot l^{\bar{a}\bar{b}} \underline{t}^{\bar{a}\bar{b}} = 5 \times 10^3 \begin{bmatrix} 2 & 1 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} \sqrt{3} \\ 1 \end{bmatrix} = 5 \times 10^3 \begin{bmatrix} 2\sqrt{3} + 1 \\ \sqrt{3} + 4 \end{bmatrix}$$

$$l^{\bar{a}\bar{b}} \underline{t}^{\bar{a}\bar{b}} \cdot \underline{\epsilon} \cdot l^{\bar{a}\bar{b}} \underline{t}^{\bar{a}\bar{b}} = [\sqrt{3}, 1] \cdot 5 \times 10^3 \begin{bmatrix} 2\sqrt{3} + 1 \\ \sqrt{3} + 4 \end{bmatrix} = 5 \times 10^3 (6 + \sqrt{3} + \sqrt{3} + 4) =$$

$$= 5 \times 10^3 (10 + 2\sqrt{3})$$

$$\frac{\underline{\underline{l^{ab}}} \cdot \underline{\underline{\epsilon}} \cdot \underline{\underline{l^{ab}}}}{\underline{\underline{l^{ab}}}} = 5 \times 10^{-3} (5 + \sqrt{3})$$

$$\underline{\underline{\epsilon}} \cdot \underline{\underline{l^{bc}}} = 5 \times 10^{-3} \cdot \begin{bmatrix} 3 \\ 5 \end{bmatrix}$$

$$\underline{\underline{l^{bc}}} \cdot \underline{\underline{\epsilon}} \cdot \underline{\underline{l^{bc}}} = 5 \times 10^{-3} \cdot (3 + 5) = 5 \times 10^{-3} \cdot 8$$

$$\frac{\underline{\underline{l^{bc}}} \cdot \underline{\underline{\epsilon}} \cdot \underline{\underline{l^{bc}}}}{\underline{\underline{l^{bc}}}} = 5 \times 10^{-3} \frac{8}{\sqrt{2}} = 5 \times 10^{-3} 4\sqrt{2}$$

$$\Delta \underline{\underline{l^{bc}}} = 5 \times 10^{-3} (5 + 4\sqrt{2} + \sqrt{3}) \approx 0,0619$$

Ejercicio 2:

a) Las tensiones obtenidas con las funciones de tensión de Airy verifican la ecuación diferencial de equilibrio.

$$\text{Con } \Phi = x^2 y \Rightarrow \tau_{xx} = 0, \quad \tau_{yy} = 2y, \quad \tau_{xy} = -2x$$

$$\frac{\partial \tau_{ij}}{\partial x_j} + \rho b_i = 0 \Rightarrow \begin{cases} \frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} = 0 + 0 = 0 \Rightarrow \text{Verifico} \\ \frac{\partial \tau_{yx}}{\partial x} + \frac{\partial \tau_{yy}}{\partial y} = -2 + 2 = 0 \Rightarrow \text{Verifico} \end{cases}$$

b)

$$\tau_{ij} = \lambda \varepsilon_{kk} \delta_{ij} + 2\mu \varepsilon_{ij} \quad (1)$$

$$\tau_{mm} = \lambda \varepsilon_{kk} \delta_{mm} + 2\mu \varepsilon_{mm} \Rightarrow \varepsilon_{kk} = \frac{\tau_{kk}}{3\lambda + 2\mu} \quad (2)$$

$$(1) \text{ y } (2) \Rightarrow \varepsilon_{ij} = \frac{1}{2\mu} \left(\tau_{ij} - \frac{\lambda}{3\lambda + 2\mu} \tau_{kk} \delta_{ij} \right)$$

$$\tau_{kk} = 2y$$

$$\varepsilon_{xx} = \frac{1}{2\mu} \left[0 - \frac{\lambda}{3\lambda + 2\mu} 2y \right] = -\frac{\lambda}{\mu(3\lambda + 2\mu)} y$$

$$\varepsilon_{yy} = \frac{1}{2\mu} \left[2y - \frac{\lambda}{3\lambda + 2\mu} 2y \right] = \left(1 - \frac{\lambda}{3\lambda + 2\mu} \right) \frac{1}{\mu} y$$

$$\varepsilon_{xy} = \frac{1}{2\mu} (-2x) = -\frac{1}{\mu} x$$

$$\left. \begin{aligned} \frac{\partial^2 \varepsilon_{xx}}{\partial y^2} &= 0 \\ \frac{\partial^2 \varepsilon_{yy}}{\partial x^2} &= 0 \\ \frac{\partial^2 \varepsilon_{xy}}{\partial x \partial y} &= 0 \end{aligned} \right\} 0 + 0 = 0 \Rightarrow \text{Verifica la ecuación de compatibilidad}$$

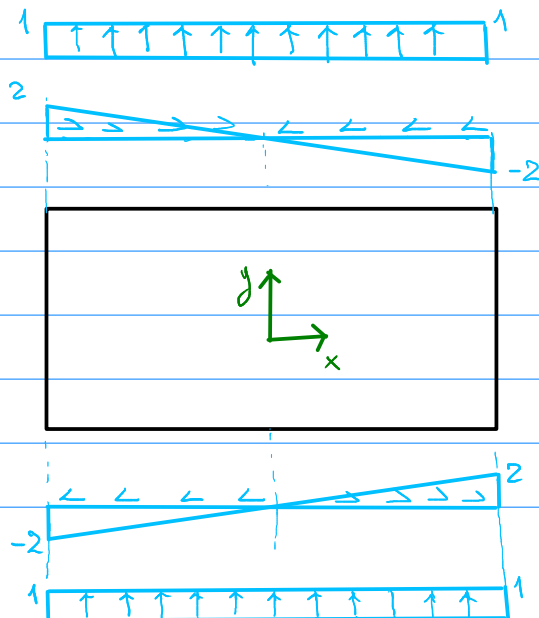
c)

lado superior e inferior:

$$\underline{T}(x, y = \pm 0,5) = \underline{T}(x, y = \pm 0,5) \cdot \begin{bmatrix} 0 \\ \pm 1 \end{bmatrix}$$

$$\underline{T} = \begin{bmatrix} 0 & -2x \\ -2x & 2y \end{bmatrix} \Rightarrow \underline{T}(x, \pm 0,5) = \begin{bmatrix} 0 & -2x \\ -2x & \pm 1 \end{bmatrix}$$

$$\underline{T}(x, \pm 0,5) = \begin{bmatrix} -2x \\ \pm 1 \end{bmatrix}$$



Resultados sobre cara superior:

$$N = \int_{x=-1}^{x=1} \int_{z=0}^{z=e} 1 \, dz \, dx = \int_{x=-1}^{x=1} 1 \, dx \cdot \int_{z=0}^e dz =$$

$$= 2 \cdot \frac{2}{e} = 4$$

(también considerando que es un carga uniforme)

$$Q = e \cdot \int_{x=-1}^{x=1} -2x \, dx = e \cdot \left. -\frac{2x^2}{2} \right|_{-1}^1 = e \cdot (1-1) = 0$$

(también considerando la antisimetría de la carga)

centro de momento, baricentro de la sección superior.

sobre la cara superior es cero

$$M_z = e \cdot \int_{x=-1}^{x=1} (x \cdot T_y - y \cdot T_x) \, dx = e \cdot \int_{x=-1}^{x=1} x T_y \, dx$$

$$= e \cdot \int_{x=-1}^{x=1} 0,5x \, dx = e \cdot \left. \frac{0,5x^2}{2} \right|_{x=-1}^{x=1} = 0$$

(también considerando que la carga uniforme no genera momento con respecto a su centro y la carga triangular de tensiones de corte mantiene brago de potencia)

Ejercicio 3:

$$A_{mnpq} = \rho_{mi} \rho_{nj} \rho_{pk} \rho_{ql} A_{ijkl}$$

$$= \rho_{mi} \rho_{nj} \rho_{pk} \rho_{ql} \delta_{ik} \delta_{jl}$$

$$= \underbrace{\rho_{mk} \rho_{nl} \rho_{pk} \rho_{ql}}_{\delta_{mp}} =$$

$$A_{mnpq} = \delta_{mp} \delta_{qn} \Rightarrow \text{los mismos componentes que } A_{ijkl} \Rightarrow$$
$$\Rightarrow \underline{\underline{A}} \text{ es isotrópico}$$

Ejercicio 4

Cantidad de movimiento lineal: $P_i = \int_V \rho v_i dV$

Resultado de las fuerzas externas: $F_i = \int_S T_i dS + \int_V b_i dV$

Fórmula de Cauchy: $T_i = n_j \tau_{ji}$

Balance de la cantidad de movimiento lineal: $\frac{D}{Dt}(P_i) = F_i$

Ecuación de continuidad: $\frac{D\rho}{Dt} + \rho \frac{\partial v_\delta}{\partial x_\delta} = 0$

Velocidad: $v_i = \frac{Du_i}{Dt}$

Tensor de pequeñas deformaciones: $\epsilon_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)$

Deformación volumétrica: $\epsilon_{KK} = \frac{\partial u_K}{\partial x_K}$

$$\int_S T_i dS = \int_S n_j \tau_{ji} dS = \int_V \frac{\partial \tau_{ji}}{\partial x_j} dV \Rightarrow$$

$$\Rightarrow F_i = \int_V \left(\frac{\partial \tau_{ji}}{\partial x_j} + b_i \right) dV$$

$$\frac{D P_i}{Dt} = \frac{D}{Dt} \left(\int_V \rho v_i dV \right) = \int_V \left[\frac{D}{Dt} (\rho v_i) + \rho v_i \frac{\partial v_\delta}{\partial x_\delta} \right] dV =$$

$$= \int_V \underbrace{\left(\frac{D\rho}{Dt} + \rho \frac{\partial v_\delta}{\partial x_\delta} \right)}_{0 \text{ por Ec. de continuidad}} v_i dV + \int_V \rho \frac{D v_i}{Dt} dV = \int_V \rho \frac{D v_i}{Dt} dV$$

0 por Ec. de continuidad

$$\frac{D}{Dt}(P_i) = f_i \Rightarrow \int_V \left[\rho \frac{Dv_i}{Dt} - \left(\frac{\partial \tau_{ji}}{\partial x_i} + b_i \right) \right] dV = 0 \quad \forall V \Rightarrow$$

$$\rho \frac{Dv_i}{Dt} = \frac{\partial \tau_{ji}}{\partial x_i} + b_i //$$

$$\tau_{ji} = \lambda \epsilon_{kk} \delta_{ji} + 2\mu \epsilon_{ji} \Rightarrow \frac{\partial \tau_{ji}}{\partial x_i} = \lambda \delta_{ji} \frac{\partial \epsilon_{kk}}{\partial x_i} + 2\mu \cdot \frac{\partial \epsilon_{ji}}{\partial x_i}$$

$$\begin{aligned} \frac{\partial \tau_{ji}}{\partial x_i} &= \lambda \frac{\partial \epsilon_{kk}}{\partial x_j} + 2\mu \cdot \frac{\partial}{\partial x_i} \left[\frac{1}{2} \left(\frac{\partial u_j}{\partial x_i} + \frac{\partial u_i}{\partial x_j} \right) \right] = \\ &= \lambda \frac{\partial \epsilon_{kk}}{\partial x_j} + \mu \left[\frac{\partial^2 u_j}{\partial x_i \partial x_i} + \frac{\partial}{\partial x_j} \underbrace{\left(\frac{\partial u_i}{\partial x_i} \right)}_{\epsilon_{ii}} \right] = \\ &= (\lambda + \mu) \cdot \frac{\partial \epsilon_{kk}}{\partial x_j} + \mu \frac{\partial^2 u_j}{\partial x_i \partial x_i} \end{aligned}$$

dunque:

$$\rho \frac{D^2 u_i}{Dt^2} = (\lambda + \mu) \cdot \frac{\partial \epsilon_{kk}}{\partial x_j} + \mu \cdot \frac{\partial^2 u_j}{\partial x_i \partial x_i} + b_i //$$

Exercício 5.

a)

$$\frac{Dq_i}{Dt} = \frac{\partial q_i}{\partial t} + \frac{\partial q_i}{\partial x_j} \cdot v_j$$

$$\frac{\partial q_i}{\partial t} = \begin{bmatrix} 2x_1 t \\ -x_2 \sin(t) \\ x_3 \cdot e^t \end{bmatrix}$$

$$\frac{\partial q_i}{\partial x_j} v_j = \begin{bmatrix} x_2 & 0 & 0 \\ 0 & \cos(t) & 0 \\ 0 & 0 & e^t \end{bmatrix} \cdot \begin{bmatrix} x_1/t \\ x_2 \cdot \frac{\sin(t)}{\cos(t)} \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 t \\ x_2 \sin(t) \\ x_3 e^t \end{bmatrix}$$

$$\frac{Dq_i}{Dt} = \begin{bmatrix} 2x_1 t + x_1 t \\ -x_2 \sin(t) + x_2 \sin(t) \\ x_3 e^t + x_3 e^t \end{bmatrix} = \begin{bmatrix} 3x_1 t \\ 0 \\ 2x_3 e^t \end{bmatrix}$$

b)

$$\int_S \frac{Dq_i}{Dt} \cdot n_i dS = \int_V \frac{\partial}{\partial x_i} \left(\frac{Dq_i}{Dt} \right) dV$$

$$\frac{\partial}{\partial x_i} \left(\frac{Dq_i}{Dt} \right) = 3t + 2e^t$$

$$\int_V \underbrace{(3t + 2e^t)}_{\text{independente de } x_i} \cdot dV = (3t + 2e^t) \cdot \underbrace{\int_V dV}_A = (3t + 2e^t) A$$